UNIT
5

COORDINATION CHEMISTRY



Alfred Werner
(1866 –1919)

Alfred Werner was a Swiss chemist who explainedthe bonding in coordination complexes. Werner proposed his coordination theory in 1893. It must be remembered that this imaginative theory was proposed before the electron had been discovered by J.J. Thompson in 1896. Werner did not have any modern instrumental techniques at his time and all his studies were made using simple reaction chemistry. Complexes must have been a complete mystery without any knowledge of bonding or structure. This theory and his painstaking work over the next 20 years won Alfred Werner the Nobel Prize for Chemistry in 1913.He was the first inorganic chemist to win the Nobel Prize.

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Learning Objectives

After studying this unit, students will be able to

- define important terms in coordination chemistry
- nomenclate the coordination compounds in accordance with the guidelines of IUPAC
- describe different types of isomerism in coordination compounds
- discuss the postulates of Werner's theory of coordination compounds
- predict the geometry of coordination compounds using valence bond theory
- apply crystal field theory to explain the colour and magnetic properties of coordination compounds
- differentiate high spin and low spin coordination compounds
- explain the stability of coordination compounds interms of stability constants.
- explain the applications of coordination compounds in day to day life

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